

Reference excerpt 02-007

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from the transverse scan of the neck, where the internal jugular vein (IJV) is clearly visible. [Sisini et al. (2015)] The cross sectional area of the IJV (CSA) is measured on each picture (sonogram) of the video clip. The sequence of measurements (data sets) is the CSA trace. Moreover, in the same studies, it was shown that acquiring the video clip of the IJV simultaneously to the ECG trace, allows to identify relationship between the two ECG and JVP traces (see Fig. 1). The non-invasive techniques used up to now, are based on the use of a microphone or a motion sensor and produce a trace which qualitatively represents the pressure in the IJV.[Pyhel(1978), Applefeld(1990), Mackenzie(1992), Mackay(1947), Kalmanson et al. (1972)] This new technique is different from those used previously in that the JVP is the instantaneous value of the CSA of the IJV. The US technique for the JVP represents an improvement from qualitative to quantitative JVP. The importance of having a quantitative JVP trace had already emerged in the past when a calibrated JVP has been used.[Pyhel(1978)] This new technique goes beyond the limit of calibration because it makes it possible to extrapolate the numeric parameters from the JVP that can be used for the clinical evaluation inter and intra-patient. This is a methodology-note that describes the innovative methods developed by the author to obtain quantitative parameters obtained by the JVP with ultrasound technique. The results reported here are only illustrative and are not produced by a dedicated experiment. 1 METHODOLOGY